Copy-ready instruction: build the results engine

VOLUME 4 / RESULTS AND LEARNING INTELLIGENCE

Emergent, implement this volume as the authoritative specification for scoring explanations, results, question review, cohort comparisons and study recommendations. Deliver all 60 numbered analyses, with graceful unavailable states when evidence is insufficient. Build working calculations and drill-through evidence before adding decorative charts. Connect this engine to the exam adapters, approved question bank and immutable attempts specified in the other volumes.

Use facts before interpretations

Provide an immediate score receipt after a submission has reached the server and the grading job has completed. Identify the attempt, paper, exam preset, scoring version, key version and submission reason. Show exact marks, full-correct, partial-credit, wrong, unanswered and excluded counts. Separate awards caused by invalid questions from ordinary earned marks. Keep results pending rather than fabricate a score while grading is unavailable.

Organize analysis into Score, Subjects, Topics, Time, Strategy, Questions, Cohort and Next Steps. Every metric card must open its supporting question list. Each card must state its denominator, evidence count, observation period and calculation version. A personal count can exist after one attempt; an interpretation of ability requires broader evidence. Do not convert thin evidence into a precise-looking strength score.

Deliver the complete review workflow

Let Tejas review every question with his final server-committed response, scoring contribution, approved answer, source-linked solution, approved tags and time evidence. Display explanations only after their release condition. Preserve the original question numbering and stable identifiers even when options were shuffled. Make the same workflow available to hundreds of thousands of learners through access-controlled, asynchronously generated reports.

Treat comparisons honestly

Label all platform comparisons as platform cohort comparisons. Never label a practice percentile as an official NTA score, national rank, intelligence percentile or admission probability. Do not estimate a question-wise ability percentile from a binary correct or wrong outcome. Instead provide cohort response statistics and, where evidence supports them, a conditional correct-response time percentile. Use the explicit cohort rules and suppression gates in this volume.

ACCEPTANCE / PRECAUTIONS

- Show a working report for every supported exam adapter.
- Reconcile the score receipt and every drill-through total.
- Leave unavailable metrics informative, with the missing evidence named.

One measurement contract and reproducible score ledger

IMPLEMENTATION / DATA MODEL

Create a typed analytics contract shared by grading, dashboards, exports and admin tools. Do not let chart components invent formulas or recalculate scores independently. Materialize a versioned report from the immutable attempt and an explicitly identified approved scoring policy. Reuse that report wherever the learner sees a result.

Store the calculation inputs

Store attempt_id, learner_id, paper_version_id, exam_adapter_version, scoring_policy_version, answer_key_version, started_at, deadline_at, submitted_at and submission_reason. For each delivered item store canonical_question_id, delivered_option_mapping, final_committed_response, commitment_sequence, evaluation_outcome, signed_score_contribution, maximum_contribution and award_reason. Keep browser-selected drafts distinct from server-committed responses. A draft shown in the browser is not evidence that an answer was accepted.

Use ordinary eligible items E for skill and completion metrics. Partition evaluated nonempty responses into full-correct C, partial-credit P and incorrect W, so attempted A = C + P + W. Represent unanswered U separately. Exclude invalid or award-only items from these partitions and show their count X. Define ordinary positive marks G, penalty magnitude L and separate award adjustment B, so total S = G - L + B.

Make context explicit

Include approved subject, chapter, concept, prerequisite and editorial difficulty identifiers, along with tag version and reviewer status. Include original first-exposure status, practice or timed mode, accommodation preset, language, solution exposure and integrity review status. Never merge these contexts silently. Multi-tagged questions may appear in several filtered views, but an overall count must deduplicate canonical items.

Attach a metric object with metric_id, value, unit, numerator, denominator, sample_count, eligibility_query_hash, evidence_state, unavailable_reason and calculation_version. Use null for undefined values. Do not serialize Infinity, NaN or a zero that really means unknown. Store raw precision and format only in the view layer; rounding must not affect grading. For sample medians, average the two middle sorted values when n is even. For quartiles, use zero-based position h = (n-1)*p and linearly interpolate its neighboring sorted values. Version this quantile method; keep percentile ranks on the separate inclusive-count convention.

Revisions and reproducibility

Keep the historical report and create a new revision when an approved key, item status or scoring policy changes. Explain every item-level delta and the total difference. Rebuild dependent cohort statistics using the revised version, then publish the report and cohort snapshot atomically. A cache containing an old key must not be mixed with a newly graded score.

ACCEPTANCE / PRECAUTIONS

- Verify E = C + P + W + U for ordinary eligible items.
- Verify S equals the sum of all item ledger contributions.
- Expose report revision time and scoring policy identifiers.

Percentiles, ties, cohort size and comparability

MATHEMATICS / PLATFORM COHORT POLICY

Implement one documented platform percentile convention: P(s) = 100 * count(score <= s) / N. Include the learner in the cohort when their attempt satisfies its eligibility rules. Give tied scores identical percentiles. Use exact decimal or rational comparisons before display rounding. This remains a platform statistic.

A testable numeric example

Use this ten-person demonstration cohort only in tests: scores 10, 20, 20, 30, 40, 40, 40, 50, 60 and 70. A score of 40 has seven scores at or below it, therefore platform percentile 70.00. A score of 20 has percentile 30.00. The highest score has 100.00. Do not display this tiny fixture as a production benchmark. The proposed production gate is at least 100 eligible distinct learners.

Specify who is being compared

Define a cohort by exact paper version, exam preset, effective key and scoring versions, mode, timer policy, solution-release exposure state and a declared date window. Use each learner's first valid attempt only when every ordinary eligible canonical item was unseen in recorded platform history before start. An encounter in another paper, any hint, or prior solution access counts as exposure and excludes the whole attempt from the primary paper cohort. Also enforce unexposed eligibility for each item benchmark. Honor declared external exposure; unreported off-platform exposure remains a limitation. Exclude unknown exposure histories; present exposed and repeat attempts separately. Do not select each learner's best result: that biases the comparison upward. Exclude staff preview accounts and confirmed invalid attempts; log every eligibility change.

Snapshot eligible N, exclusions, date window, update time and cohort query hash. Keep a running cohort visibly provisional until its window closes. If N is below 100, show personal results and an insufficient-cohort message. More learners do not repair a nonrepresentative sample; never extrapolate platform results to all Indian examinees.

Separate official results and question statistics

Official JEE Main NTA percentiles use the relevant official shift population; the total percentile is not the mean of subject percentiles, and percentile is different from percentage marks [S4-01]. Do not recreate official normalization from this platform's participants. Do not merge different mock papers into a national percentile.

For one binary-scored item show the proportion of eligible learners who answered fully correctly, including unanswered learners in the denominator. Call this a cohort response statistic. A correct learner belongs to a large tie group; that observation does not locate their individual ability. Keep conditional time comparisons separate and exclude partial-credit items from binary-only claims.

ACCEPTANCE / PRECAUTIONS

- Test ties, all-equal scores, negative scores and zero eligible learners.
- Ensure total percentile is calculated directly from total scores.
- Show N and cohort identity beside every comparison.

Question review, approved solutions and release controls

CONTENT / EXPLANATION UX

Build a complete question review workspace that preserves source fidelity and supports both text and image-based questions. Students must understand the answer, their own response and the scoring outcome without opening a separate PDF for every question. Approved source PDF crops must remain available as a fidelity fallback.

The review panel contract

Display original question number, canonical identifier, subject and approved tags; the complete stem, figures, passages and options; the learner's final committed response; accepted answer or answer set; signed marks; and the grading explanation. Show partial credit, no credit, penalty, invalid item and special award as distinct states. For numerical questions explain the configured accepted values and precision rules rather than quietly rounding the student response.

Use the canonical option identity when explaining a shuffled option. If the learner saw option B but the master PDF calls the same answer D, show both mappings clearly. Store original question PDF page and bounding box, original solution PDF page and mapping evidence. Offer a report-content-problem control attached to those identifiers, without changing the learner's saved response.

Make solutions trustworthy

Prefer the owner-provided approved solution. Permit an additional explanation only after subject review; label its origin and version. Do not let an AI silently replace the supplied answer key, invent a derivation or fix an ambiguous question. Offer step-by-step text, source images, accessible mathematics and an optional alternate method. Keep a distinct correction notice when a supplied solution has been reviewed and amended.

Use approved error notes for common distractors and numerical traps. Explain exactly which assumption or step is wrong. A response alone seldom proves the student's reasoning, so phrase the note as a possible misconception and let the learner confirm or reject it. Do not call every wrong answer careless.

Enforce release on the server

Support immediate post-submission release, scheduled release after a test window, and teacher-controlled release. Before release, return no answer key, hidden solution text, distractor explanation or predictable asset URL in the API, page source, preload or export. Publish an explicit release timestamp and show a pending explanation state. Snapshot solution exposure so later attempts are correctly labeled as exposed practice.

Keep marks available when scoring is complete even if explanations are embargoed, subject to the test owner's configured result-release policy. Never let cohort statistics leak a delayed answer through an option marked correct.

ACCEPTANCE / PRECAUTIONS

- Test embargoed APIs and PDF downloads using a student account.
- Check every solution mapping against its canonical question.
- Preserve source accessibility and zoom on small screens.

A01-A05: score and response completeness

ANALYSIS CATALOG / 01 OF 12

Implement these five measurements from the authoritative item ledger. Permit exact personal facts after one completed attempt, while keeping teaching interpretations cautious. All counts refer to ordinary eligible items unless the card explicitly includes excluded or award-only items. Let every count open a filtered question list.

A01 / Net awarded marks

Inputs: signed contributions under the current approved key. Formula: S = sum(score_q), including explicitly identified awards. Evidence: a finalized grading revision; otherwise show grading pending. Action: inspect the largest lost-mark contributors and any adjustment notice. Limit: raw marks compare directly only within the same scoring context; they do not indicate national rank or percentile.

A02 / Fully correct response count

Inputs: final committed response and full-correct evaluation outcome. Formula: C = count(full_correct). Evidence: all relevant items graded; preserve a pending count if any remain unresolved. Action: revisit correct answers whose reasoning felt uncertain. Limit: an observed correct answer can result from guessing or prior exposure and does not establish concept mastery.

A03 / Incorrect and partial outcomes

Inputs: distinct evaluation outcomes. Formula: W = count(incorrect), P = count(partial_credit); display both separately within this analysis. Evidence: approved adapter classification. Action: open the wrong list first, then examine why partial answers failed full-credit criteria. Limit: a partially correct multiselect response is not equivalent to a wrong single-choice answer; do not flatten their explanations.

A04 / Attempted-answer accuracy

Inputs: C and A = C + P + W. Formula: 100 * C / A. Evidence: A > 0; show the exact fraction for any A, but suppress strength labels below 10 attempts. Action: inspect inaccurate attempted topics before increasing attempts blindly. Limit: excluding unanswered items can make selective students look stronger; pair this value with completion and partial-credit counts.

A05 / Response completion rate

Inputs: attempted A and eligible delivered E. Formula: 100 * A / E, with unanswered count E - A. Evidence: E > 0 and submission finalized. Action: distinguish unvisited questions from viewed but unanswered questions before planning coverage. Limit: completion rewards submitting a response, not correct reasoning; high completion can coexist with negative marks and poor accuracy.

ACCEPTANCE / PRECAUTIONS

- Return null when A or E is zero.
- Never count an unsaved browser draft as an attempt.
- Keep excluded-item awards visible outside ordinary outcome counts.

A06-A10: marks earned, penalties and subject balance

ANALYSIS CATALOG / 02 OF 12

Implement score decomposition without inventing a single strategy score. Read the signed item contributions produced by each exam adapter; question categories may carry different rewards, penalties and partial-credit rules. Present values in marks first and make all ratios explicit about their denominator.

A06 / Ordinary positive marks earned

Inputs: ordinary item contributions s_q. Formula: G = sum(max(s_q, 0)), excluding separate award-only adjustments B. Evidence: finalized ordinary grading. Action: identify which subjects and question types generate the positive marks. Limit: G ignores penalties and is not the final result; partial-credit contributions remain positive marks rather than full-correct counts.

A07 / Penalty marks lost

Inputs: negative ordinary contributions. Formula: L = sum(max(-s_q, 0)). Evidence: finalized grading; zero is valid when the preset has no negative marking. Action: review the exact penalized responses and their stated confidence where available. Limit: a penalty does not prove an irresponsible guess; a student may have made a sincere reasoning error.

A08 / Penalty-to-positive-marks ratio

Inputs: L and G. Formula: 100 * L / G. Evidence: G > 0; otherwise show L with ratio unavailable. Action: compare the ratio across comparable first attempts and inspect preventable error categories. Limit: values may exceed 100 percent; do not clamp them. This is not penalty divided by final net marks, which becomes unstable near zero.

A09 / Percentage of configured maximum marks

Inputs: total S and current effective maximum M from the scoring policy. Formula: 100 * S / M. Evidence: M > 0 and all special awards resolved. Action: read alongside raw marks to understand the scale of this paper. Limit: allow negative percentages where rules permit negative scores; this percentage is not a cohort percentile or an official normalized score.

A10 / Subject marks contribution

Inputs: item-to-subject assignment and signed contributions. Formula: S_subject = sum(s_q within subject); show G_subject, L_subject and awards separately. Evidence: every item has one approved primary scoring subject. Action: choose a subject for detailed review based on its actual mark losses and coverage. Limit: subjects with different item counts or time demands cannot be judged solely by raw totals.

ACCEPTANCE / PRECAUTIONS

- Reconcile subject sums exactly to the total score.
- Test zero gross marks with nonzero penalties.
- Display different marking categories rather than assuming one reward/penalty pair.

A11-A15: chapters, concepts, response types and exposure

ANALYSIS CATALOG / 03 OF 12

Implement academic breakdowns from reviewed tags. Display descriptive group counts whenever available, but require at least 10 eligible attempted items before showing performance-oriented recommendations for a group. Identify insufficient coverage instead of presenting one question as a reliable chapter judgment. All thresholds here are proposed product defaults, not validated ability cutoffs.

A11 / Chapter mark contribution

Inputs: approved chapter tags, item score and tag weights. Formula: sum(w_qg * s_q), where weights across a question's assigned chapters sum to one. Evidence: reviewed chapter mapping; show recommendations only for groups with at least 10 attempts. Action: locate chapters contributing losses. Limit: fractional attribution is an accounting convention, not a measurement of each chapter's causal role.

A12 / Concept attempted accuracy

Inputs: fully correct and attempted items matching one approved concept. Formula: 100 * C_concept / A_concept. Evidence: denominator positive; advice requires at least 10 distinct items across two sessions. Action: inspect repeated reasoning errors within the concept. Limit: overlapping tags make separate concept cards nonadditive, and unequal item difficulty can distort comparisons.

A13 / Response-type performance

Inputs: response type, signed marks and configured maxima. Formula: sum(s_q) / sum(max_q), separately for single-choice, numerical and supported multiselect categories. Evidence: positive maximum; recommend changes only above 10 items per type. Action: compare handling of input and selection tasks. Limit: response-type differences may reflect topic and marking differences rather than input skill.

A14 / Editorial difficulty outcome matrix

Inputs: human-approved difficulty labels and final outcomes. Formula: C/A and net marks within each editorial band, with unanswered counts. Evidence: at least 10 attempted items per band for interpretation; unknown labels remain a separate row. Action: identify easier-band errors worth checking first. Limit: editorial difficulty is a judgment and must not masquerade as nationally calibrated difficulty.

A15 / First-exposure versus repeat performance

Inputs: canonical item exposure history and solution access before the attempt. Formula: C/A separately for unseen, previously seen and solution-exposed items. Evidence: known exposure history and at least 10 attempts per compared group. Action: favor fresh questions when checking transfer. Limit: remembered answers and self-selected repeats inflate performance; cross-group gaps do not estimate the effect of studying.

ACCEPTANCE / PRECAUTIONS

- Deduplicate items within each filtered concept view.
- Display unknown tags and exposure status explicitly.
- Keep weighted chapter totals additive and concept totals clearly overlapping.

A16-A20: timer use and observable question time

ANALYSIS CATALOG / 04 OF 12

Collect low-cost visibility and visit events, then distinguish official timer consumption from observed on-screen activity. Reading can occur without interaction, and missing telemetry is not inactivity. Define a reconciled timing ledger before deriving time advice. Suppress question-time advice below 90 percent usable timing coverage for the relevant group.

A16 / Timer time consumed

Inputs: server start, effective deadline, authorized adjustments and submission receipt. Formula: min(submitted_at, deadline_at) - started_at, adjusted by documented timer rules. Evidence: complete server clock ledger. Action: examine whether substantial available time remained at submission. Limit: elapsed time includes offline or hidden-page periods in an unpaused timed exam; it does not measure active thinking.

A17 / Observed visible question time

Inputs: question-enter, question-leave, visibility and capped heartbeat intervals. Formula: union duration of valid visible intervals attributed to each question, summed without overlap. Evidence: valid timestamp order and coverage flags. Action: compare where observable attention was allocated. Limit: visibility does not establish mental focus, and a static visible page may still be actively read.

A18 / Unobserved timer fraction

Inputs: timer duration D, visible attributed T and recorded hidden duration H. Formula: 100 * max(D - T - H, 0) / D after interval reconciliation. Evidence: D > 0; distinguish known hidden time from missing events. Action: inspect connection or telemetry gaps before trusting timing advice. Limit: unobserved time must not be called laziness, cheating or distraction.

A19 / Time on slow incorrect responses

Inputs: wrong items, observed time and approved per-item correct-response median. Formula: sum(T_q where incorrect and T_q > 2 * benchmark_q). Evidence: five usable wrong items and at least 30 correct cohort times per benchmark. Action: review stopping and revisiting tactics. Limit: this descriptive threshold is configurable; slow errors may reflect legitimate effort, not wasted thought.

A20 / Question budget overrun

Inputs: a student-approved pretest time budget b_q and observed T_q. Formula: sum(max(T_q - b_q, 0)); list individual overruns. Evidence: budget existed before the attempt and valid timing; otherwise unavailable. Action: adjust the next practice budget after reviewing difficulty. Limit: the overrun is measured against a plan, not an optimal schedule or guaranteed lost marks.

ACCEPTANCE / PRECAUTIONS

- Do not double-count simultaneous tabs or duplicated heartbeats.
- Test reconnection, backgrounding and early submission.
- Keep the server deadline independent of the analytics heartbeat stream.

A21-A25: allocation, revisits and answer changes

ANALYSIS CATALOG / 05 OF 12

Implement these analyses from ordered visits and accepted response commitments. A navigation click is not an answer change, and a local selection may never become a saved response. Show the actual sequence and policy-scored contributions when explaining changes. Use observed time only where its coverage is adequate.

A21 / Subject time allocation

Inputs: valid question-visible time and primary subject. Formula: 100 * T_subject / sum(T_subject), alongside absolute minutes. Evidence: positive observed time and at least 90 percent usable item timing. Action: compare allocation with the student's pretest plan and mark contribution. Limit: equal subject time is not inherently desirable; different papers and personal strengths require different allocations.

A22 / First-visit duration

Inputs: each eligible item's first valid visible visit. Formula: median(first_visit_seconds), with a distribution by outcome. Evidence: at least five usable first visits per reported group. Action: identify where an initial scan became an extended attempt. Limit: first-visit time may include reading shared passages or diagrams; it is not pure solution time.

A23 / Revisit time share

Inputs: visible visits ordered per item. Formula: 100 * sum(duration of visits after the first) / total attributed time. Evidence: positive total and at least five visited items. Action: inspect whether revisits resolved uncertainty or repeated an unproductive approach. Limit: a high revisit share can be a sensible checking strategy; it is not automatically inefficient.

A24 / Positive score difference after revisions

Inputs: first committed response, final response and current scoring policy. Formula: sum(max(score(final_q) - score(first_commit_q), 0)). Evidence: complete commitments for revised items; show exact differences after one revision. Action: review the reasoning behind beneficial changes. Limit: this compares recorded endpoints and cannot prove that changing answers caused improvement or predict future gains.

A25 / Negative score difference after revisions

Inputs: the same committed endpoints. Formula: sum(max(score(first_commit_q) - score(final_q), 0)). Evidence: complete revision history; annotate clears and timeout acceptance rules. Action: study harmful changes and the evidence used to justify them. Limit: final correctness is hindsight; do not prescribe never changing answers from one outcome or count uncommitted drafts as lost marks.

ACCEPTANCE / PRECAUTIONS

- Render first and final canonical option identities consistently.
- Use the current key for both revision endpoints.
- Show unknown revision history instead of silently assuming no changes.

A26-A30: late responses, review recovery and omissions

ANALYSIS CATALOG / 06 OF 12

Implement a descriptive view of how the attempt developed over time. Do not infer fatigue, stress, attention disorders or cheating from these observations. Divide the attempt using its authorized duration and keep the relevant timestamp definition visible. Tie every strategy suggestion to actual question evidence and the learner's stated experience.

A26 / Late-window attempted accuracy

Inputs: time of first accepted response commitment and final outcome. Formula: C/A for items first committed in the final 20 percent of authorized duration versus earlier items. Evidence: at least 10 items in each group and valid timestamps. Action: review late errors and pace planning. Limit: late items may be harder or from another subject; the gap does not diagnose fatigue.

A27 / Recovery of unanswered review items

Inputs: items first marked for review while unanswered and their final outcomes. Formula: recovered full-correct items / initially unanswered review items. Evidence: at least five such items and ordered review events. Action: examine which revisit choices converted omissions into correct responses. Limit: a review flag is an interface state, and successful recovery does not reveal the optimal review order.

A28 / Unvisited eligible-item rate

Inputs: delivered ordinary items and successful question-render events. Formula: 100 * count(no render recorded) / E. Evidence: E > 0, reconciled navigation events and at least 95 percent timer duration accounted for by known visible or hidden intervals. Action: practice an initial scan if entire sections were missed. Limit: missing render events can mimic unvisited questions; show unavailable when navigation evidence is incomplete.

A29 / Time spent on viewed omissions

Inputs: final unanswered items with recorded visits and visible duration. Formula: sum(T_q for viewed and unanswered items), with count and question list. Evidence: valid item timings and at least 90 percent relevant coverage. Action: review whether a timely skip or an alternate method would help. Limit: unanswered effort may improve understanding; do not automatically classify these minutes as waste.

A30 / Median time on fully correct items

Inputs: visible attributed durations for fully correct responses. Formula: median(T_q), split by subject or response type. Evidence: at least five correct timed items per personal group. Action: identify unusually long correct solutions to compare with an approved shorter method. Limit: faster is not always better, and comparing unlike item mixtures can give a misleading speed conclusion.

ACCEPTANCE / PRECAUTIONS

- Keep late-window definitions stable across report revisions.
- Exclude invalid items from omission measures.
- Provide descriptive wording instead of psychological labels.

A31-A35: honest cohort response comparisons

ANALYSIS CATALOG / 07 OF 12

Use the same frozen cohort specification for all five analyses on this page. The proposed primary gate is 100 eligible distinct learners on the exact paper and scoring context. Display eligibility, counts, snapshot time and whether the window is still open. Never convert these results into an official entrance-exam rank.

A31 / Overall platform cohort percentile

Inputs: the learner's net score s and eligible cohort scores. Formula: 100 * count(score <= s) / N. Evidence: N >= 100; otherwise suppress the percentile. Action: compare standing on this specific mock while reviewing subject gaps. Limit: this self-selected platform cohort is not the official exam population; an apparent rise can reflect cohort changes.

A32 / Subject platform cohort percentile

Inputs: approved subject net scores for the same cohort. Formula: 100 * count(subject_score <= learner_subject_score) / N. Evidence: N >= 100 and common subject scoring. Action: identify subjects with weaker relative standing on this paper. Limit: never average subject percentiles to produce total percentile; ties and differing distributions prevent that shortcut.

A33 / Item full-correct cohort proportion

Inputs: one binary-eligible item's full-correct outcomes and all eligible delivered attempts. Formula: p_q = C_q / N_q, including unanswered responses in N_q. Evidence: N_q >= 100. Action: check wrong answers on widely solved questions first. Limit: this is a cohort response statistic, not the student's question-wise ability percentile or a permanent intrinsic difficulty rating.

A34 / Item accuracy among attempted responses

Inputs: the same item's full-correct and nonempty committed responses. Formula: C_q / A_q. Evidence: cohort N_q >= 100 and A_q >= 30. Action: distinguish low success from low willingness to attempt. Limit: students select which questions to attempt, so this conditional rate is biased toward respondents and should always accompany the overall proportion.

A35 / Committed option-selection distribution

Inputs: canonical options chosen in a single-choice item. Formula: count(option_j) / A_q for every option, with unanswered displayed separately. Evidence: N_q >= 100 and A_q >= 30; suppress option cells below 10 learners in shared views. Action: examine common distractors after solution release. Limit: popularity alone cannot establish the underlying misconception, and shuffled option letters must be canonicalized.

ACCEPTANCE / PRECAUTIONS

- Suppress correct-option disclosure while solutions are embargoed.
- Include unanswered learners in the overall item proportion.
- Do not show decimal precision unsupported by cohort size.

A36-A40: item timing and content quality signals

ANALYSIS CATALOG / 08 OF 12

Implement conditional timing comparisons only when the learner answered fully correctly and the benchmark has adequate usable times. Restrict comparison to matching timer policies and exposure states. Treat item quality signals as review triggers for teachers, not automatic decisions to delete content or accusations against learners.

A36 / Correct-response time percentile

Inputs: the learner's visible time t and valid correct cohort times. Formula: 100 * count(cohort_time >= t) / N_correct; higher means faster-or-equal within correct responses. Evidence: cohort >= 100, N_correct >= 30 and valid personal time. Action: consider a shorter approved method. Limit: correctness-conditioned selection and telemetry uncertainty prevent an ability-percentile interpretation.

A37 / Cohort median correct-response time

Inputs: valid times from fully correct, first-exposure cohort responses. Formula: median(times), with the interquartile range. Evidence: cohort >= 100 and at least 30 usable correct times. Action: compare expected practice pacing with this specific question's observed distribution. Limit: the median reflects this cohort and interface; it is not an official ideal time or universal time limit.

A38 / Personal-to-benchmark time ratio

Inputs: valid personal time t and positive A37 median m. Formula: t / m. Evidence: A37 available and learner fully correct; otherwise provide personal time only. Action: inspect ratios substantially above a configurable threshold without chasing speed indiscriminately. Limit: reading accommodations, language and solution style can affect timing; the ratio does not measure intelligence or concentration.

A39 / Item omission cohort rate

Inputs: eligible delivered item attempts and final unanswered outcomes. Formula: U_q / N_q. Evidence: N_q >= 100; distinguish known unvisited from viewed omissions only with usable telemetry. Action: recognize questions the cohort often postpones. Limit: omissions can reflect placement, section strategy or time pressure as well as difficulty; do not infer a single cause.

A40 / Item upper-minus-lower response difference

Inputs: binary outcomes and rest-of-test net scores excluding this item. Formula: p_upper - p_lower for highest and lowest rest-score quartiles. Evidence: N >= 200, at least 50 per group and nonconstant rest scores; resolve boundary ties with documented fractional weights. Action: flag negative or unusual values for content review. Limit: this descriptive discrimination signal is unstable under narrow score ranges and is not proof of a bad question.

ACCEPTANCE / PRECAUTIONS

- State the reverse-time percentile convention in the tooltip.
- Suppress benchmark timing if correct-time evidence is thin.
- Route item quality flags through human review.

A41-A45: confidence, expected value and error attribution

ANALYSIS CATALOG / 09 OF 12

Collect confidence only in optional practice features or post-test reflection; never add compulsory controls to an official-format simulation. A recalled confidence rating must be labeled retrospective and excluded from pre-response calibration. Keep student assumptions distinct from measured outcomes and teacher-confirmed explanations. These analyses describe evidence rather than predicting admissions.

A41 / Confidence calibration by bin

Inputs: optional precommit probability ratings p and binary outcomes y. Formula: mean(p) and mean(y) within declared probability bins. Evidence: at least 20 rated items overall and 10 per displayed bin. Action: inspect confident errors and cautious successes. Limit: self-reported probabilities are noisy; observed calibration does not establish a stable personal trait or justify universal guessing rules.

A42 / Binary Brier score

Inputs: genuine precommit p in [0,1] and y equal to one for full correctness. Formula: mean((p - y)^2), with lower values indicating closer predictions. Evidence: at least 20 binary-rated distinct items; omit partial-credit items. Action: improve probability estimates using reviewed explanations. Limit: item mix affects the score, and this is calibration evidence, not an intelligence or mastery score.

A43 / Low-confidence penalty exposure

Inputs: precommit confidence, configurable threshold c and signed contributions. Formula: sum(max(-s_q,0) where p_q < c), plus the rated attempt count. Evidence: at least 10 genuinely rated attempts; default c = 0.20 is a product proposal. Action: review these choices against the marking policy. Limit: low confidence does not prove blind guessing, and retrospective confidence is excluded.

A44 / Assumption-based expected marks

Inputs: user-supplied correctness probability p, full reward a and wrong penalty magnitude b for a simple binary item. Formula: EV = p*a - (1-p)*b; break-even p = b/(a+b). Evidence: a+b > 0 and compatible marking rules. Action: explore practice decisions. Limit: this is a hypothetical model; unknown p and time opportunity cost prevent guaranteed gains.

A45 / Confirmed error-cause distribution

Inputs: learner-confirmed or teacher-reviewed causes for wrong items. Formula: count(cause) / number of labeled wrong items, with unknown and unlabeled counts separate. Evidence: at least five labeled errors before recommending a pattern. Action: select a targeted checking drill. Limit: a chosen distractor does not prove a cause; preserve multiple possible causes and allow labels to be corrected.

ACCEPTANCE / PRECAUTIONS

- Exclude retrospective ratings from predictive calibration.
- Validate probabilities and marking compatibility.
- Never infer an error cause solely from an option letter.

A46-A50: progress, variability and practice backlog

ANALYSIS CATALOG / 10 OF 12

Provide longitudinal analyses only after enough comparable evidence exists. Different papers cannot be assumed equivalent just because they share an exam name. Show the blueprint, exposure mix, maximum marks and mode next to every comparison. Prefer new questions when checking learning and label repeated-question effects clearly.

A46 / Comparable first-attempt score change

Inputs: latest score percentage and the preceding three comparable first-attempt score percentages. Formula: latest minus median(previous three). Evidence: four approved blueprint-matched tests with matching mode and scoring context. Action: inspect which topic outcomes accompany the change. Limit: blueprint similarity does not guarantee equal difficulty; the difference is descriptive rather than a causal learning estimate.

A47 / Fresh concept accuracy change

Inputs: distinct new concept-tagged items in two chronological windows. Formula: 100 * (C_recent/A_recent - C_prior/A_prior). Evidence: at least 20 items in each window, reviewed tags and no solution exposure. Action: verify whether the concept needs another drill. Limit: changing difficulty and item selection can explain the gap; show counts and overlap checks.

A48 / Comparable score variability

Inputs: score percentages from comparable first attempts. Formula: sample standard deviation sqrt(sum((x-mean)^2)/(n-1)). Evidence: at least five tests; display the individual observations. Action: review large swings and the associated exam conditions. Limit: variability is not a personality, anxiety or attention diagnosis, and small test histories cannot establish a stable reliability profile.

A49 / Repeated-item score difference

Inputs: first and later score for the same canonical item under the same effective key. Formula: mean(later_score - first_score) across paired items. Evidence: at least 10 pairs and known exposure state. Action: use improvement as feedback, then test a fresh related item. Limit: remembered answers can drive the gain; do not mix repeated-item performance into first-exposure cohort percentiles.

A50 / Unresolved review backlog

Inputs: wrong or omitted first-exposure items and explicit review-task states. Formula: count(open review tasks), with age since the attempt and topic breakdown. Evidence: at least one qualifying task; no mastery label. Action: schedule a manageable next review block. Limit: closing a task records workflow completion, not concept mastery; automatic closure requires a clearly stated fresh-item check policy.

ACCEPTANCE / PRECAUTIONS

- Show paper comparability details before a progress chart.
- Do not select each student's best attempt as the baseline.
- Keep completed workflow and demonstrated transfer separate.

A51-A55: recurring errors, transfer checks and review priorities

ANALYSIS CATALOG / 11 OF 12

Build these measures as evidence-linked study aids. Use reviewed concept and prerequisite tags, deduplicate repeated canonical items and retain uncertainty. A topic becomes a review priority because specific recent responses justify it, not because an opaque algorithm assigns a frightening weakness percentage. Let learners edit their priorities.

A51 / Recurring concept error count

Inputs: distinct incorrect first-exposure items sharing a reviewed concept across sessions. Formula: count(distinct wrong canonical items), with attempted denominator. Evidence: at least five attempted concept items over three sessions before a recurring-pattern flag. Action: revisit the underlying method rather than memorizing individual answers. Limit: related items and difficulty differences prevent treating the count as an independent diagnosis.

A52 / Fresh review-check success rate

Inputs: new, approved counterpart items attempted after a documented review task. Formula: full-correct checks / attempted fresh checks. Evidence: at least 10 checks; exclude the original item and solution-exposed items. Action: decide whether to continue the drill or expand to harder examples. Limit: selected checks can be easier than exams; success demonstrates these checks, not guaranteed transfer.

A53 / Delayed fresh-check accuracy

Inputs: fresh counterpart items taken 14 days after review, allowing a declared plus/minus seven-day window. Formula: C_delayed/A_delayed. Evidence: at least 10 fresh checks with valid review and attempt dates. Action: schedule spaced review if errors recur. Limit: this proposed schedule is configurable and later accuracy also reflects additional learning, item mix and missing follow-up participation.

A54 / Assigned prerequisite coverage

Inputs: teacher-approved prerequisite tags and qualifying attempted practice. Formula: covered assigned tags / total assigned prerequisite tags; define covered as at least one valid attempt. Evidence: an explicit assignment and a positive tag count. Action: identify prerequisites not yet practiced. Limit: coverage records exposure, not understanding; one attempt cannot certify readiness for a dependent concept.

A55 / Error-category marks-at-stake ceiling

Inputs: confirmed error-category items, current score s_q and item maximum m_q. Formula: sum(m_q - s_q) for those wrong items. Evidence: reviewed cause labels and valid maxima; exact count always visible. Action: prioritize categories with several concrete examples. Limit: this is a hindsight arithmetic ceiling, not predicted recoverable marks; fixing every error may require time and may not recur.

ACCEPTANCE / PRECAUTIONS

- Use fresh counterpart items for transfer checks.
- Label coverage separately from mastery.
- Do not present arithmetic ceilings as expected future score gains.

A56-A60: solution engagement, revisions and evidence quality

ANALYSIS CATALOG / 12 OF 12

Implement these final measurements to explain what the platform knows about review activity and report reliability. Keep them visible in the evidence drawer as well as the study workflow. They must never be folded into a hidden ability score. Missing content or telemetry should reduce claims, not punish the learner.

A56 / Released-solution consultation rate

Inputs: eligible released review items and authorized solution-open events. Formula: distinct authorized opened solutions within the eligible review set / distinct released review items. Evidence: a positive released-item denominator and valid event access. Action: open relevant explanations for unreviewed errors. Limit: opening a solution does not prove reading or understanding; no engagement event should be collected before the solution becomes accessible.

A57 / Observed solution review duration

Inputs: visible solution-panel visits and capped heartbeat intervals. Formula: union of observed visible durations, with median duration per opened item. Evidence: at least five usable solution visits for a summary. Action: compare a review routine with its planned duration. Limit: time on screen is not learning time, and copying a method is not demonstrated transfer.

A58 / Approved-key score revision impact

Inputs: preserved historical report and newly approved grading revision. Formula: S_new - S_old, with every changed item contribution listed. Evidence: both reproducible ledgers and a published correction reason. Action: understand the corrected result and refreshed comparisons. Limit: score changes from a key revision are not improvement or deterioration in the learner's ability.

A59 / Approved explanation availability

Inputs: released delivered items and approved question-to-solution mappings. Formula: mapped approved explanations / released delivered items, with unavailable identifiers. Evidence: positive released-item denominator; count only approved mappings. Action: report missing explanations and revisit when corrected. Limit: an available mapping does not guarantee explanatory quality, so preserve source review status and student feedback.

A60 / Usable timing evidence coverage

Inputs: eligible items, reconciled visit events and timing quality flags. Formula: items with usable timing / eligible items; also show known timed duration / timer duration. Evidence: positive denominators and the declared quality policy. Action: understand why some speed suggestions are unavailable. Limit: high coverage does not prove precise thinking time; missing events and visible-but-unfocused periods remain possible.

ACCEPTANCE / PRECAUTIONS

- Release-gate all solution activity and availability calculations.
- Exclude key revisions from student progress signals.
- Expose evidence-quality limitations beside derived advice.

Question-wise risks, precautions and expected-value reasoning

INTERPRETATION / EXPLAINABLE STRATEGY

Generate each question's risk and precaution notes from approved content plus recorded evidence. Separate a content trap, an input-format trap and a decision tradeoff. Give the learner a practical next check. Never replace a reviewed explanation with a generic warning or imply that a probability model knows what the student truly understood.

Use a disciplined note template

For every flagged question show: observation, supporting evidence, possible explanation, next precaution and limitation. Example: "Your final response has the wrong sign. The solution changes sign when the integration limits are reversed. Check the limit order before simplifying. This response alone cannot prove where your reasoning diverged." Attach the exact solution step or reviewed content note. Let the learner label their cause after reading.

Support reviewed traps such as units, domain restrictions, extraneous roots, sign conventions, numerical formatting, option quantifiers and multiselect completeness. Display only notes relevant to the approved item and its adapter. A blank response gets a coverage or decision note, not an invented misconception. Correct but slow items may receive an alternate-method suggestion; correct but low-confidence items may receive a reasoning check.

Explain the model rather than prescribing guesses

For a simple +4/-1 item, EV = 4p - (1-p) = 5p - 1, so the arithmetic break-even probability is 0.20. A hypothetical 0.25 probability gives EV 0.25 marks per attempt. A +1/0 policy gives EV p when blank scores zero. These are examples of configured rules, not assumptions that all exams use them.

Explain that p is generally unknown. Elimination of options improves a theoretical probability only if the eliminated options really cannot be correct. Time has an opportunity cost, and the remaining test may contain better uses of that time. Multiselect, partial-credit, bonus and numerical-response rules need their own outcome models; disable the binary calculator when those assumptions fail.

Offer simulation as post-test reflection

Permit students to try a transparent hindsight scenario, such as removing selected low-confidence penalized answers. List changed items and recompute with the exact adapter. Label the result "hypothetical score on this attempt." Do not call it a guaranteed attainable score. A hindsight scenario must not alter the saved attempt, published cohort or official-format result.

ACCEPTANCE / PRECAUTIONS

- Require reviewed source evidence for each content trap.
- Keep hypothetical probabilities and observed accuracy distinct.
- Display scenario assumptions and preserve original results.

Tags, evidence summaries and an editable study plan

LEARNING WORKFLOW / NO OPAQUE MASTERY SCORE

Create a reviewed academic taxonomy that supports searching, analytics and study planning. Make topic conclusions explainable in ordinary language. Use evidence states such as insufficient evidence, mixed recent outcomes and consistently correct recent practice. Treat these as descriptive summaries of selected evidence, not precise estimates of a student's total knowledge.

Build and review the taxonomy

Use subject, chapter, concept, prerequisite, response type and editorial difficulty as separate dimensions. Attach syllabus version, owner, reviewer and review date to each academic tag. Let AI suggest tags during ingestion, but require approval before tags drive student recommendations. Keep uncertain tags out of automatic prerequisite chains. Preserve redirects when taxonomy names change so historical reports remain interpretable.

Allow multiple concepts for one item and make the evidence traceable. Never sum overlapping concept counts to produce a total. Use the fractional chapter attribution rule only when additive totals are required, and explain it in the report. A learner's notes and confirmed error causes are personal annotations, distinct from the published question taxonomy.

Use explicit recommendation rules

Propose at most three study actions per report. Prioritize a concept when fresh incorrect items recur across sessions, the tags are reviewed and the sample gate is satisfied. If evidence is thin, suggest a short diagnostic practice set instead of assigning a weakness label. Explain the rule in the card: "Three sessions include errors on five distinct questions tagged with this concept." Link the exact questions.

Combine a short source-linked review, an approved fresh-item check and a planned follow-up window. Let students choose available time and move tasks. Respect declared exam dates and workload without inventing a timetable from absent information. Do not claim that a fixed number of minutes guarantees a percentile gain.

Track workflow and verify transfer

Provide statuses planned, reviewed, fresh-check attempted and follow-up completed. Distinguish closing a task from demonstrating transfer. Use fresh counterpart items for a check, record solution exposure and retain an unresolved state when performance remains mixed. Ask for optional reflection on the cause of an error, with "not sure" always available.

Keep teacher recommendations editable and reversible. A teacher may override a recommendation with a reason, but cannot silently rewrite past responses or label students with medical or psychological conditions. Make study priorities useful without public shame, streak punishment or frightening rankings.

ACCEPTANCE / PRECAUTIONS

- Explain every recommended task with specific evidence.
- Keep AI-suggested tags in review until approved.
- Use fresh items before claiming improvement beyond memorization.

Dashboard composition, comparison views and exports

PRODUCT / RESULTS INTERFACE

Build a calm blue-themed results workspace with progressive disclosure. The first screen must answer what happened, where marks went and what to do next. Show five meaningful headline facts, then allow deeper inspection. Do not overwhelm students with 60 simultaneous cards or use decorative graphics that obscure the question evidence.

A readable first screen

Place net marks, configured maximum, attempted accuracy, completion and platform cohort percentile at the top. If the percentile is suppressed, replace its space with a clear cohort-status message. Show the grading revision and release status nearby. Use tabs or anchored sections for Score, Subjects, Topics, Time, Strategy, Questions and Next Steps; preserve filters when navigating.

Use bar charts for subject marks and topic outcomes, a timeline for observed activity, and a sortable question table for detailed review. Pair every chart with a table and textual explanation. Use color plus labels and icons for outcomes. Ensure charts, mathematical explanations, tooltips and tables work with keyboard navigation and screen readers. Keep a compact phone view without hiding essential denominators.

Support precise questions

Let students filter by subject, chapter, concept, outcome, response type, confidence availability, time band, review status and source exposure. Enable sorting by marks lost, original order, observed time and approved topic. Keep the original order one click away. Compare two compatible attempts with side-by-side contexts and question lists; warn about differing blueprints before drawing a trend.

Provide an evidence drawer for each metric with inputs, formula, denominator, sample gate and limitations. A visible unavailable value should explain how evidence can be obtained, such as taking more fresh concept questions or waiting for the cohort window to close. Do not imply that unavailable metrics are a premium upsell.

Generate useful portable reports

Offer an access-controlled PDF report, question-wise CSV and personal JSON export. Include the attempt receipt, selected charts, all available analysis summaries, unavailable reasons, next tasks and correction notices. A longer question review appendix may be optional. Use job queues, signed download links and an expiry policy. Never place a student's phone number or email in the filename.

Include report revision, cohort snapshot and generation time in each export. Reuse the same calculation output as the dashboard. For delayed solutions, generate a result-only report first and regenerate the explanation appendix after release. Avoid embedding embargoed answer data even when visually hidden.

ACCEPTANCE / PRECAUTIONS

- Verify mobile readability and keyboard drill-through.
- Reconcile exported values with the on-screen report.
- Test report downloads for authorization and embargo rules.

Teacher and administrator cohort insights

OPERATIONS / PRIVACY-PRESERVING CONTENT REVIEW

Create teacher and admin views that answer which content needs review, which assigned concepts need instruction and whether reports are current. Do not expose individual student records through broad aggregate permissions. Keep student access, teaching-group access, content-editor access and platform analytics access as separate roles.

Aggregate questions and approved teaching groups

Provide distributions of net marks, completion, subject contribution, item response rates and usable timing coverage. Show the same denominator and version metadata used in student reports. Let authorized teachers see their assigned learners and aggregate class evidence; platform-wide cohorts remain aggregate unless an explicit role grants individual access. Avoid sensitive demographic comparisons by default.

Use the proposed shared-cell privacy minimum of 10 distinct learners. Suppress complementary cells when a visible total would reveal a small hidden count by subtraction. Do not allow arbitrary filters or downloadable raw tables to bypass suppression. Product statistical gates still apply: a privacy-safe cell of 10 is not enough for a published cohort percentile requiring 100.

Review content signals before taking action

Queue items with negative upper-minus-lower differences, unusual distractor patterns, frequent source complaints or missing solution mappings. Show item version, approved tags, source crops, effective answer key, cohort sizes and exposure mix. Let a reviewer record whether the signal reflects a content defect, cohort mix or legitimate difficulty. Never automatically change a key solely because many students chose another answer.

Support a controlled correction workflow: proposal, subject review, approved revision, impact preview and publish. Show the count of affected attempts and the expected score changes before committing. Preserve the old report and issue an intelligible correction notice. Recompute item rates, cohort scores, percentiles and any study-plan facts derived from the old outcome.

Operate large cohorts without stale comparisons

Aggregate eligible first attempts asynchronously after grading. Partition by paper and cohort identity, update materialized snapshots and stamp every view with freshness. Make the scoring service authoritative and keep analytics refreshes idempotent. Monitor queue age, failed jobs, key-version mismatches and suppression counts. Use a report status that distinguishes score ready, cohort updating and explanations pending.

Expose an internal audit view of calculation definitions and changes. Teachers may annotate interpretations but must not edit formulas ad hoc. A report should remain reproducible after a cohort closes, even when later repeat attempts arrive.

ACCEPTANCE / PRECAUTIONS

- Test complementary suppression and filter abuse.
- Keep content corrections separate from statistical flags.
- Recompute every dependent metric after approved key changes.

Telemetry uncertainty, privacy and statistical safeguards

QUALITY / RESPONSIBLE MEASUREMENT

Collect the smallest event stream that supports the promised analyses. Explain what is observed and what remains unknown. Configure retention, deletion and access policies before launch, then verify them against applicable law and the platform's actual audience. Do not add camera surveillance, biometric inference or psychological profiling merely to enrich analytics.

Bound the meaning of time evidence

Record question entry and exit, accepted answer commitments, review-state changes, visibility changes and a modest capped heartbeat. Use monotonic client intervals and server-receipt validation; reconcile clock changes, offline batches and duplicate event identifiers. Do not record raw keystrokes, clipboard contents, microphone streams or unrelated browsing. A lost heartbeat should reduce confidence in timing advice, not invalidate an otherwise valid saved response.

Flag implausible interval order, excessive overlap, missing closure and unknown visibility. Cap attributed intervals according to a documented policy, then retain an uncertainty flag rather than reconstructing precise unseen activity. Keep timing analytics separate from the authoritative deadline and submission logic. A student who loses connectivity may still have timer time consumed and incomplete analytics.

Quantify uncertainty without pretending representativeness

For binary cohort proportions, display counts and a Wilson 95 percent interval when the independence approximation is reasonable. With p = k/n and z = 1.96, center = (p + z^2/(2n))/(1 + z^2/n); half-width = z*sqrt(p*(1-p)/n + z^2/(4n^2))/(1 + z^2/n). Bounds are center plus or minus half-width [S4-02].

Label this interval as conditional on the observed eligible sample. It does not correct self-selection, repeated exposure, shared coaching or correlated items. Do not show this formula as an ability interval for a single learner. Do not manufacture a percentile confidence band from arbitrary point adjustments. Keep all minimum sample gates configurable and versioned; they are product safeguards, not scientific guarantees.

Protect the learner's evidence

Keep personal responses, timing, confidence and error notes private by default. Provide an understandable explanation of optional event collection and a route to inspect personal records. Restrict sharing to authorized teachers or user-approved links. Minimize personally identifying fields in exports and aggregates. Apply deletion to source events, derived personal reports and cached downloads according to the retention policy.

Use human review for integrity flags and show limitations. A tab switch, fast response or missing event is not proof of cheating. Keep adverse decisions outside the analytics engine unless a separately approved process provides evidence and appeal.

ACCEPTANCE / PRECAUTIONS

- Verify deletion propagation into derived personal records.
- Check confidence intervals against known numerical fixtures.
- Keep missing-data states separate from zero-valued measurements.

Acceptance fixtures and final analytics release gate

VALIDATION / MUST PASS BEFORE LAUNCH

Emergent, deliver deterministic fixtures and automated checks for arithmetic, cohort eligibility, report revisions, release controls and incomplete evidence. Use the approved adapters from the exam-engine volume. A beautiful dashboard is incomplete until every displayed number is reproducible from the stored attempt and the same value appears in its export.

Test arithmetic with explicit expected values

Create a simple five-item +4/-1 fixture with three fully correct, one wrong and one unanswered response: G = 12, L = 1, S = 11, A = 4, accuracy = 75 percent, completion = 80 percent, maximum M = 20 and score percentage = 55 percent. Penalty-to-positive ratio must be 100/12 = 8.3333 percent. Add a no-negative preset, a partial-credit preset and an invalid-item award case.

Use the ten-score demonstration from the percentile page: score 40 gives 70.00 and score 20 gives 30.00 under inclusive ties. Assert that the production sample gate suppresses both. Use a 100-person binary item fixture with 40 correct, 20 wrong and 40 unanswered: overall correct proportion 40 percent, attempted accuracy 40/60 = 66.6667 percent, and omission rate 40 percent. Test an all-equal cohort and an empty cohort.

Test event and version edge cases

Test selected-but-uncommitted options, commits arriving after the deadline, clear-response events, shuffled options, duplicate messages, offline telemetry batches, hidden tabs and missing visit closures. Verify that analytics never decides which late response was accepted; the exam engine's committed final state does. Validate zero denominators, negative marks, no usable correct times and incomplete exposure histories.

Change one approved key in a fixture and assert that historical and revised ledgers coexist, score deltas reconcile, dependent cohort statistics refresh and progress trends exclude correction-caused gains. Assert that a stale cache cannot combine a revised score with an old percentile.

Test student-facing safety and completeness

Attempt to obtain embargoed answers through report APIs, preloaded assets, exported CSV, cached PDF, option statistics and shared links. Verify role restrictions, privacy suppression and complement suppression. Check that every A01-A60 has an implemented formula, evidence gate, drill-through, explanation and unavailable state. Show at least 40 distinct substantive analysis types in a fully populated demonstration history, without duplicating labels to inflate the count.

Finish with subject-review checks of question explanations and tags, accessibility checks of charts and tables, and export reconciliation. Publish the analytics calculation registry and the proposed product thresholds with their versions. Keep launch blocked on failed scoring, access-control or explanation-release checks.

ACCEPTANCE / PRECAUTIONS

- Deliver fixture inputs, expected outputs and reproducible test commands.
- Check all 60 catalog entries against working screens.
- Provide a concise report of passed checks and unresolved limitations.

Source register and implementation handoff

EVIDENCE AND SCOPE OF THIS SPECIFICATION

Use these references to verify the stated baseline. Requirements and engineering thresholds in this volume are proposed product decisions unless a source explicitly establishes them.

S4-01 | National Testing Agency: JEE Main 2026 Information Bulletin, Appendix VII

https://cdnbbsr.s3waas.gov.in/s3f8e59f4b2fe7c5705bf878bbd494ccdf/uploads/2025/11/20251102 1649722475.pdf

Checked: 2026-10-11. Primary official source for the distinction between raw marks, official shift-based percentile normalization and subject-versus-total percentiles. This volume proposes separate platform cohort calculations; recheck the official bulletin for the actual target exam year.

S4-02 | NIST/SEMATECH e-Handbook: 7.2.4.1 Confidence intervals for proportions

https://www.itl.nist.gov/div898/handbook/prc/section2/prc241.htm

Checked: 2026-10-11. Primary statistical reference for the Wilson interval formula. The interval does not solve platform cohort selection bias or establish learner ability; sample gates in this volume are proposed product policy.

ACCEPTANCE / PRECAUTIONS

- Before release, retain the approved configuration, source document date, verification owner and test evidence. Re-check any rule that changes for a new exam year.